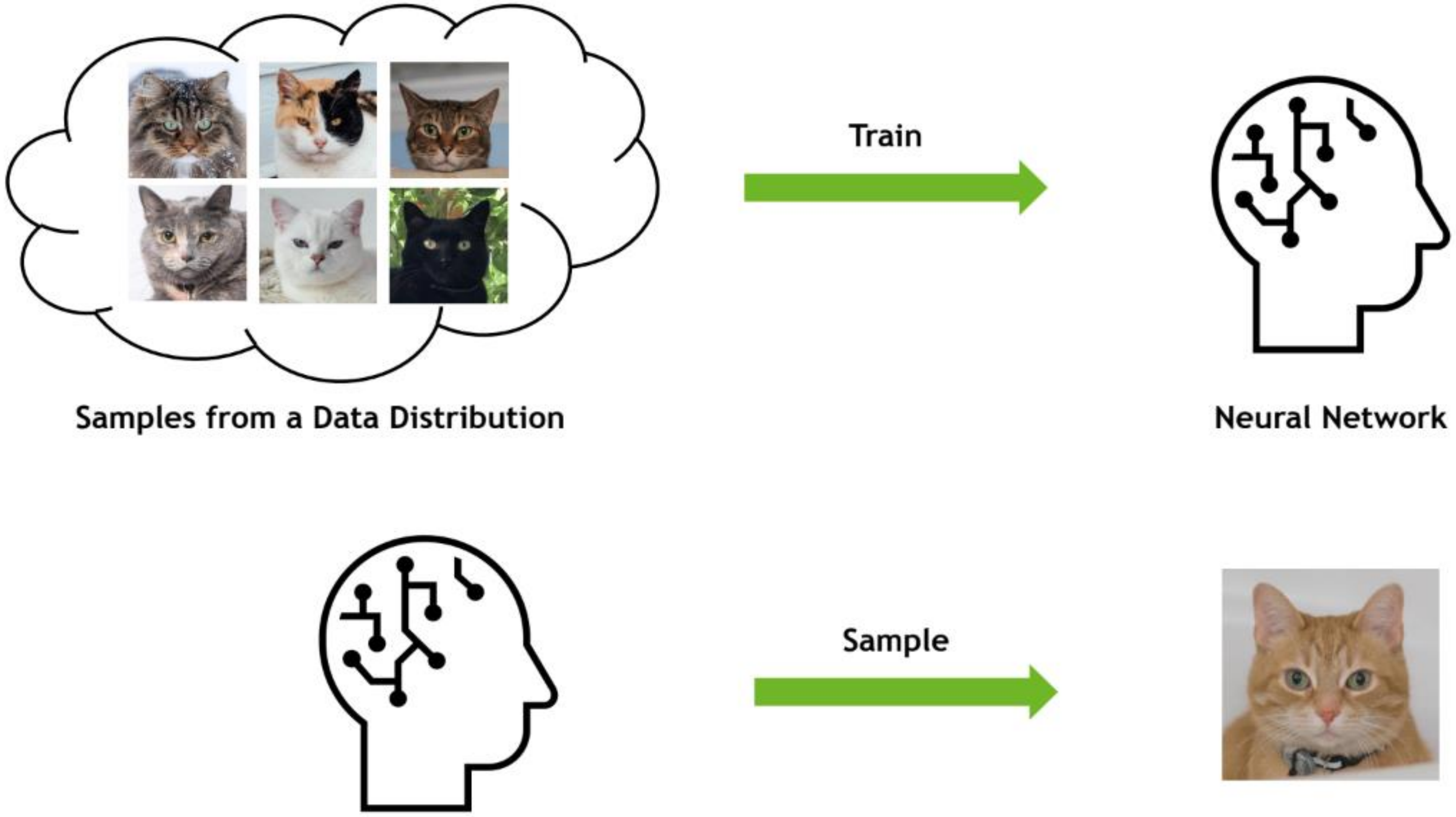
Computer Vision and Deep Learning

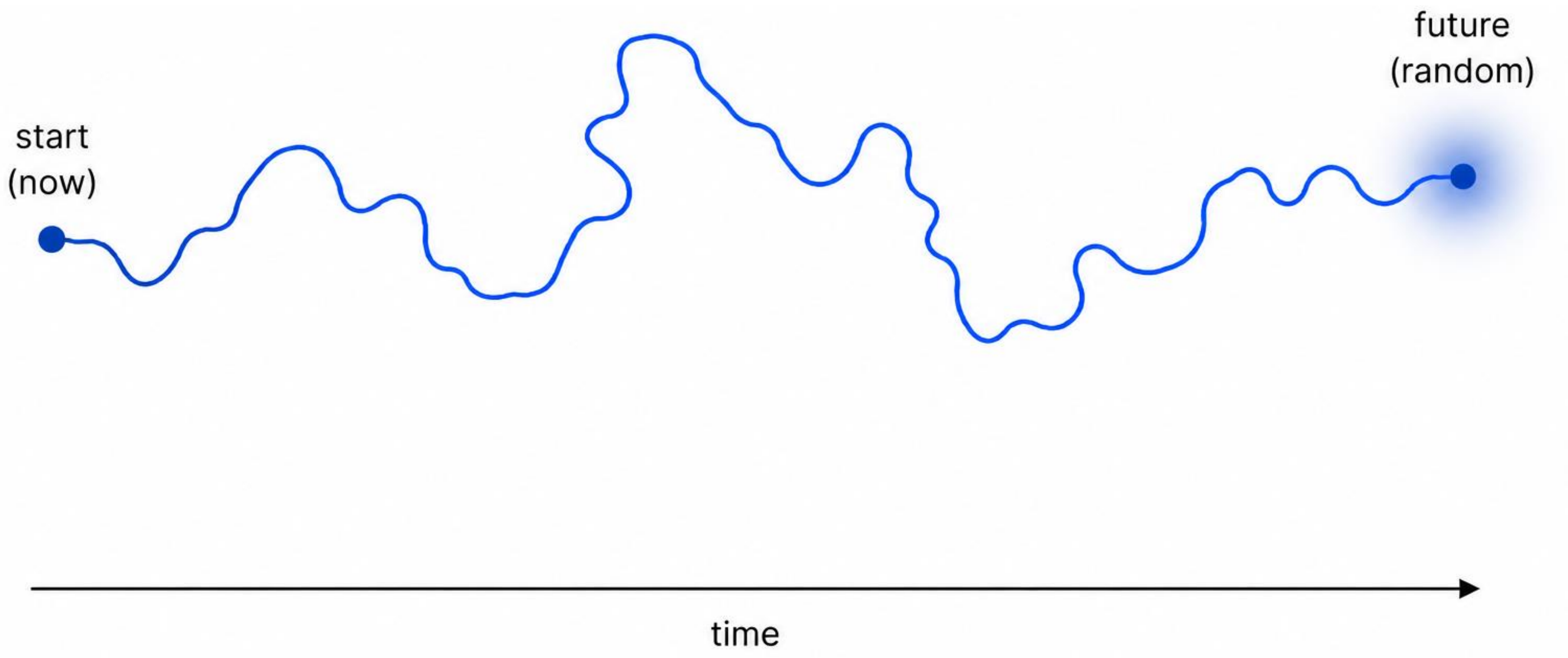
Lab 07: Diffusion Models

Background: Deep Generative Learning

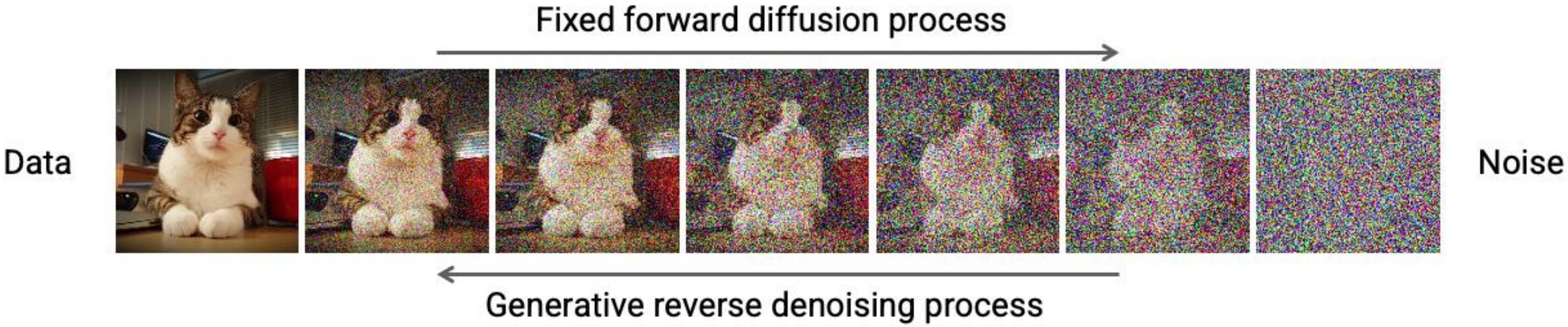


Background: Diffusion Process

- A diffusion process is a stochastic Markov process having continuous sample paths.



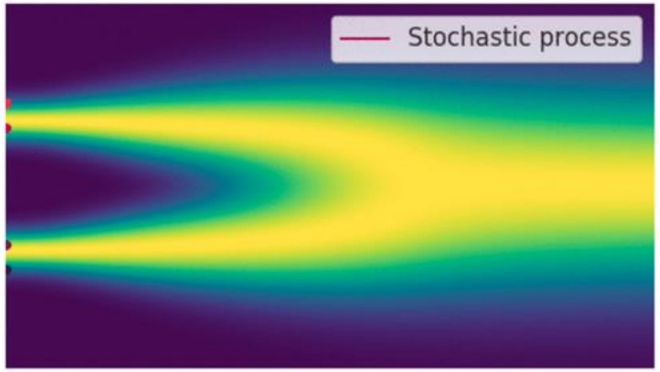
Denoising Diffusion Probabilistic Model (DDPM)



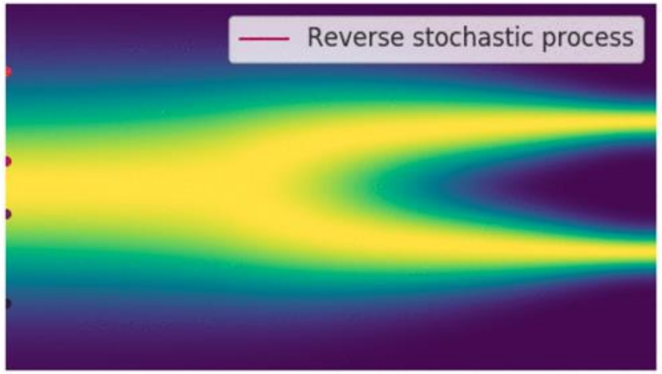
DDPM consists of two processes:

1. Forward diffusion process gradually adds noise to the input
2. Reverse denoising process learns to generate data by denoising

Denoising Diffusion Probabilistic Model (DDPM)



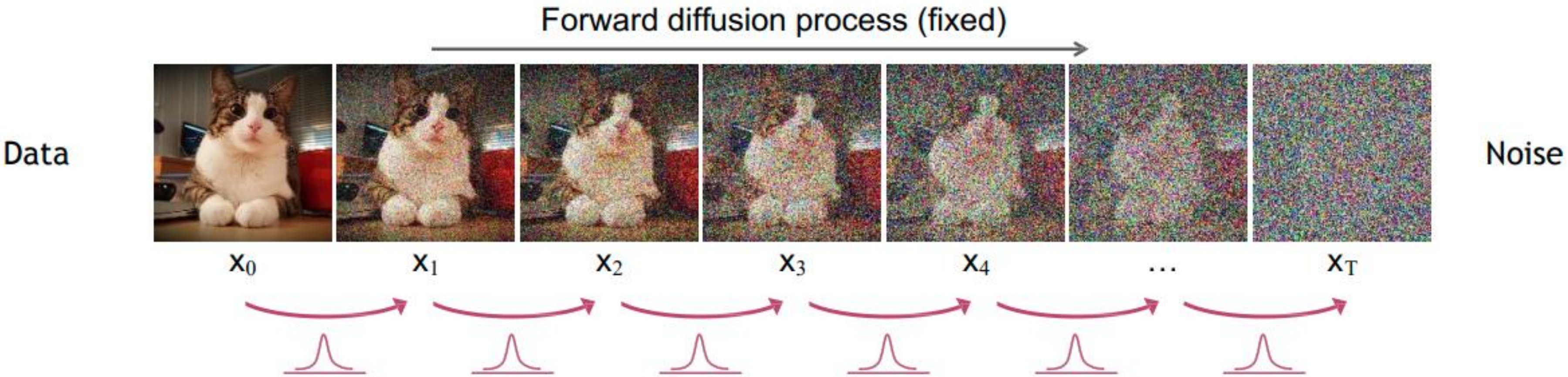
- "Destroy" the data by gradually adding small amounts of gaussian noise



- "Create" data by gradually denoising a noisy code from a stationary distribution

DDPM- Forward Process

The formal definition of the forward process in T steps:



$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x_t}; \sqrt{1-\beta_t}\mathbf{x_{t-1}}, \beta_t\mathbf{I}) \quad \Rightarrow \quad q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}) \quad \text{(joint)}$$

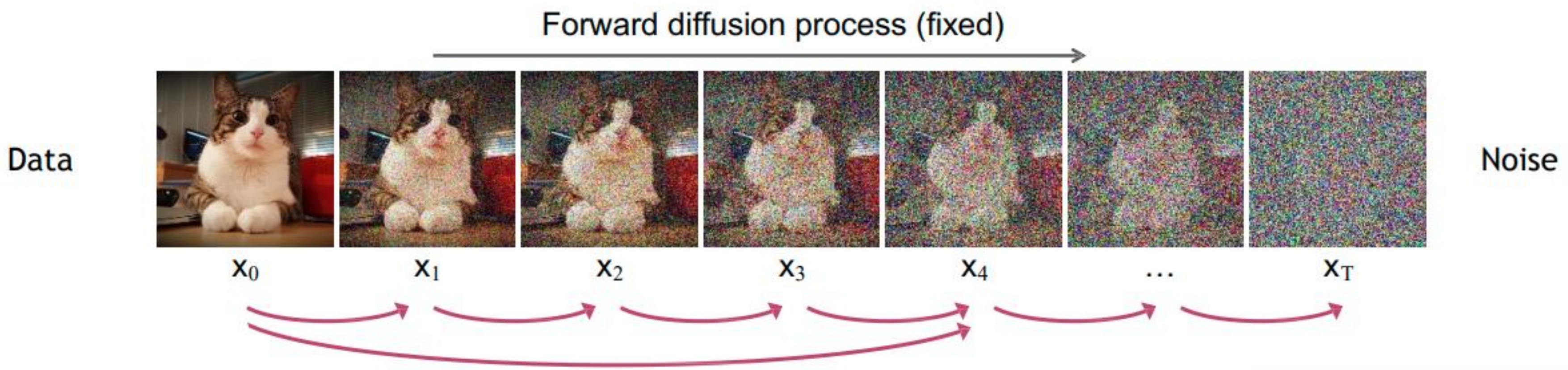
Image at prev. Step
(less noisy)

Mean
Mostly keep prev. Image
Slightly shrink (so image slowly fades)

Variance
(noise strength)

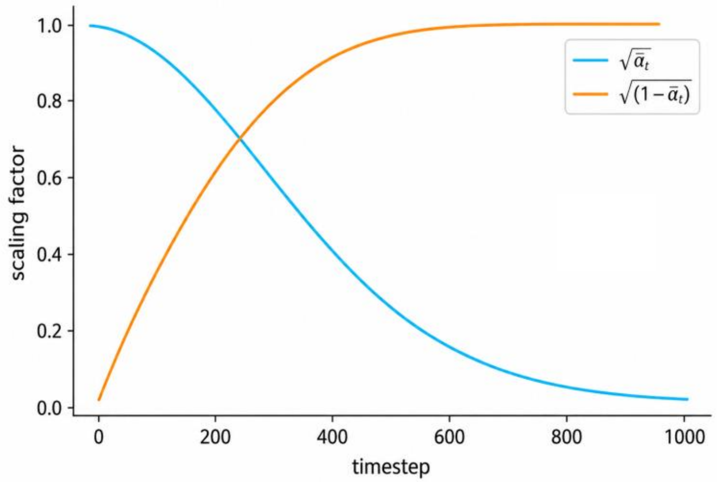
Probability distribution of of
the whole seq. x1, …, xT
given the starting image x0

DDPM- Forward Process

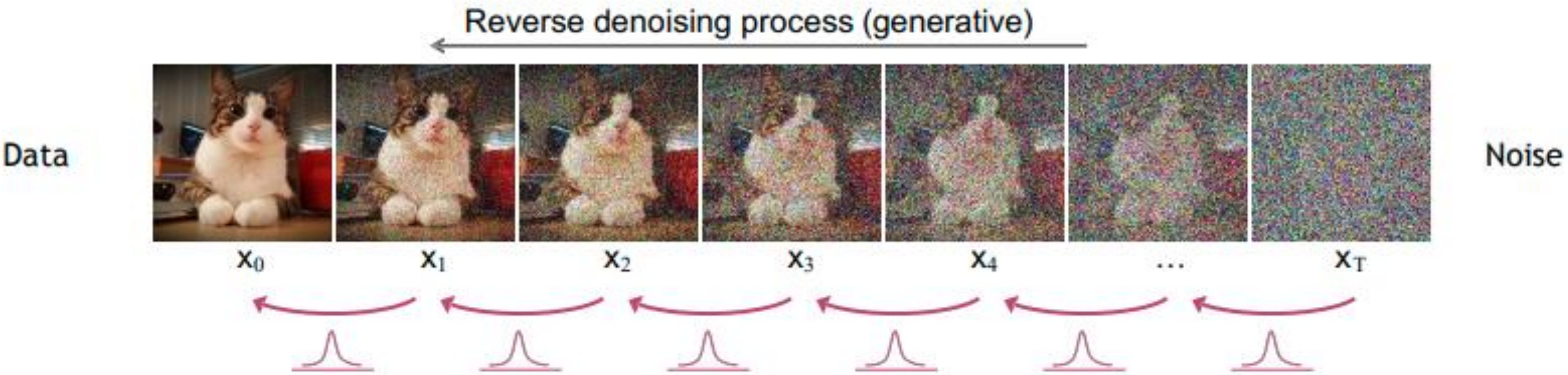


Define $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$ ➡ $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}))$

For sampling: $\mathbf{x}_t = \underbrace{\sqrt{\bar{\alpha}_t}\, \mathbf{x}_0}_{\text{Signal / Image part}} + \underbrace{\sqrt{(1-\bar{\alpha}_t)}\, \epsilon}_{\text{Noise part}}$ where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$



DDPM- Reverse Process



$$p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$$

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boxed{\mu_\theta(\mathbf{x}_t, t)}, \boxed{\sigma_t^2}\mathbf{I})$$

$$p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

Given a noisy image xt, predict a slightly less noised version xt-1

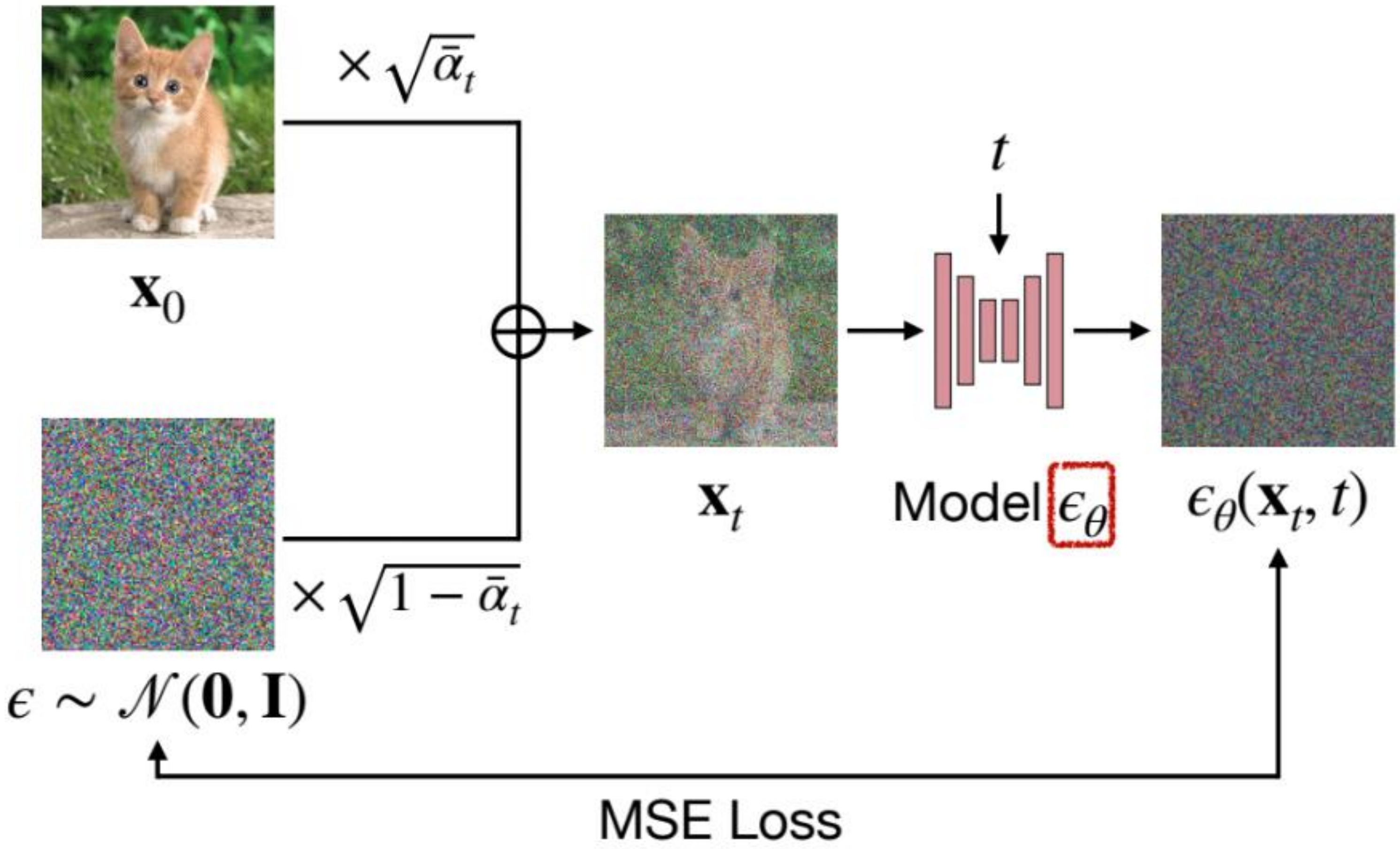
Parametric model

The model estimates the expected (mean) value of xt-1 given xt

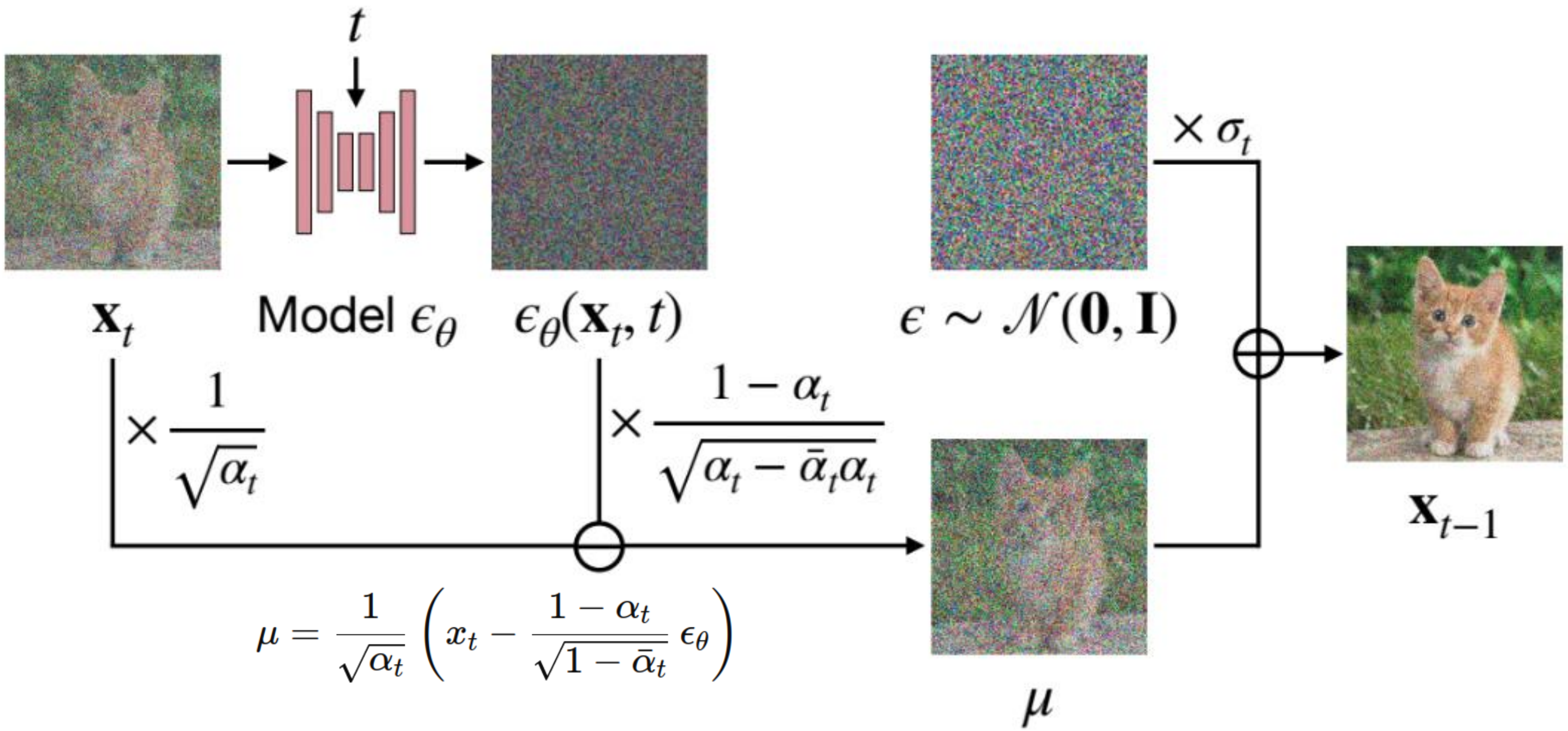
i.e. Beta_t

Probability of an entire trajectory from noise -> image

DDPM- Training

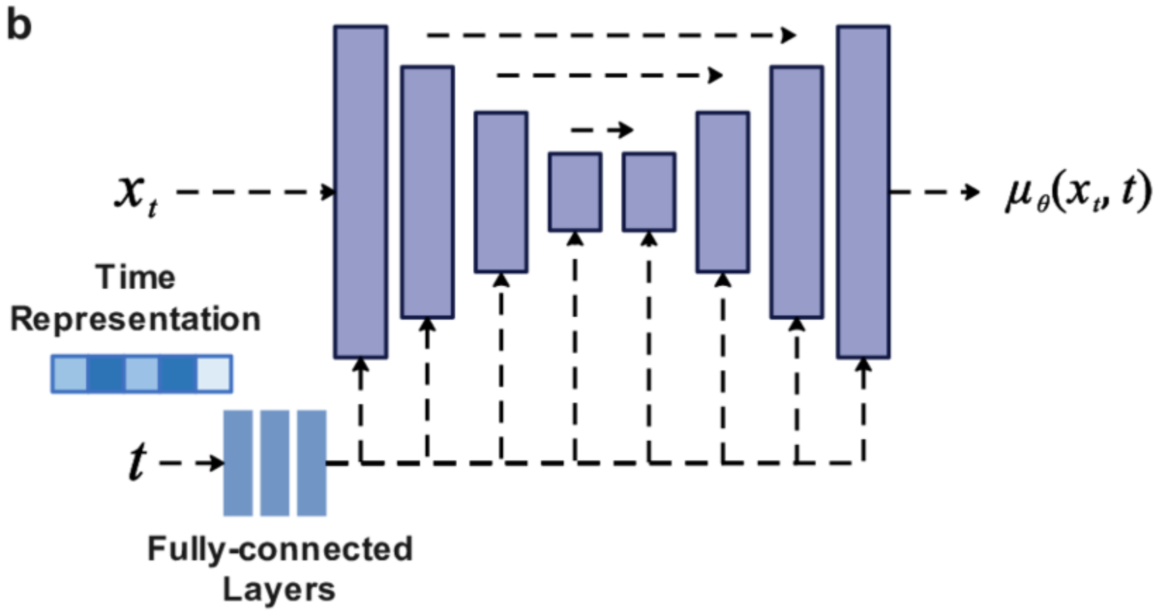


DDPM- Sampling

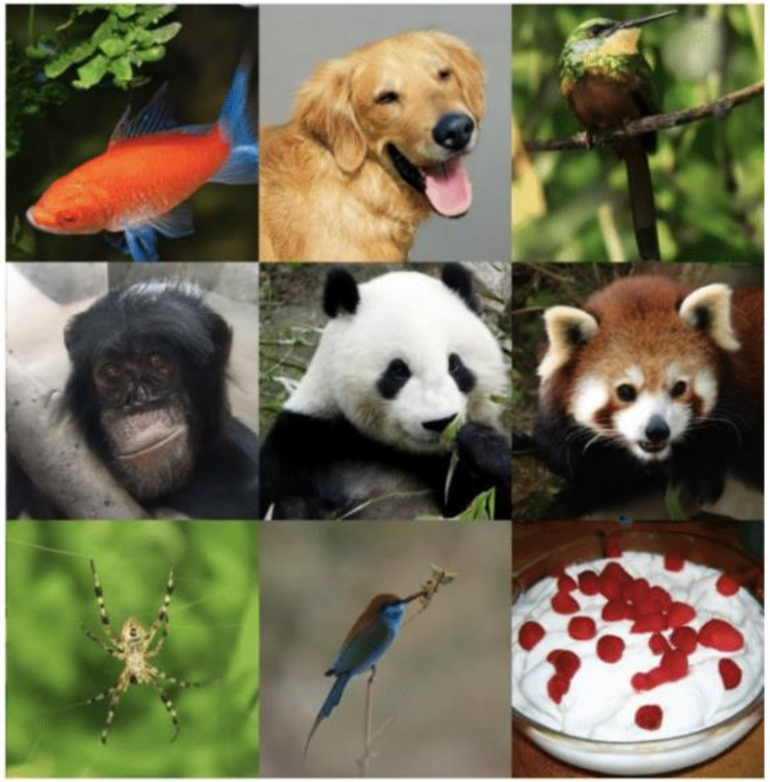


(best estimate of the
next denoised image)

DDPM- UNet


b
x_t
$\mu_\theta(x_t, t)$
Time
Representation
t
Fully-connected
Layers

DDPM- Results



"Diffusion Models Beat GANs on Image Synthesis"
Dhariwal & Nichol, OpenAI, 2021

"Cascaded Diffusion Models for High Fidelity Image Generation"
Ho et al., Google, 2021

⚗ Exercises